Influence of Feature Importance on the Performance of Machine Learning Models trained on k-Anonymous Data

Chanh Minh Tran[1,a)] Kazuhiro Minami[1,b)]

Abstract: Enforcing k-anonymity on a dataset often results in information loss as original values of data features are generalized, perturbed, or even suppressed. However, existing works have shown inconsistent consequences of such information loss on the performance of machine learning models trained on different k-anonymous datasets compared to those trained on the original versions. Specifically, with large values of k and high information loss, some datasets suffer from severe deterioration in machine learning performance, while some datasets can provide similar performance. In this work, we investigate the influence of feature importance in addition to information loss. Experimental results demonstrate that machine learning performance do not significantly deteriorate as long as information loss on the important features is low.

Keywords: k-anonymity, information loss, classification, feature importance

1. Introduction

k-anonymity has been widely studied as a fundamental privacy-preserving criterion to facilitate safe micro-data publishing. It guarantees any record in a data table is indistinguishable from at least $k-1$ other records with respect to a *quasi-identifier* (QID), i.e., set of attributes that may enable the re-identification of individuals [1]. This is often implemented by ensuring every query on any QID results in an *equivalence class* (E) of at least k records. Given an original data D and a privacy parameter k, an algorithm is called k-anonymization if it takes D and k as inputs and outputs an anonymized data D' satisfying k-anonymity.

In the literature, numerous algorithms have been proposed for k-anonymization [2], [3], [4], [5], following several common approaches, such as generalization [2], [3] or local recoding [4], [5]. Regardless of algorithms or approaches, enforcing k-anonymity usually causes a degree of information loss on the anonymous data compared to original data, as some data values are altered during the anonymization process. This potentially leads to the degradation in performance of downstream data analysis task, such as classification, because the data becomes less detailed or noisy. Thus, in order to evaluate a k-anonymization algorithm, one usually relies on data utility metrics that estimate the information loss incurred on anonymous data [2], [5], [6].

In case of classification, intuitively, classifiers trained on anonymous data with higher information loss should provide lower classification performance. In our previous work [7], we showed that the utility metric called *Normalized Certainty Penalty* (NCP) intuitively expressed the information loss and could correlate properly with classification performance on one dataset. However, when comparing across different datasets, our follow-up work [8] concluded that the correlation was inconsistent. Specifically, we showed that datasets suffered from information loss (measured by NCP) differently; while classification performance on one dataset (HOUSING [9]) was highly sensitive to information loss, another dataset (ADULT [10]) showed only minor performance degradation even with severe information loss. This hints that characteristics of the dataset itself also influence the classification performance, in addition to information loss on its anonymous version.

Meanwhile, a line of research in k-anonymization considers the *feature importance* of the QID attribute as one of the inputs for anonymization [11], [12]. As each feature (QID attribute) may contribute differently to the classification result, these work propose to prioritize anonymization on those less important features, while avoiding anonymization as much as possible on the important ones. Inspired by these work, in this paper, we investigate the potential influence of feature importance as a utility metric, in addition to information loss, on subsequent classification performance of anonymous data. We found out that, even with high overall information loss, if the exclusive information loss on the important features are low, classification performance is not significantly affected.

[1] Research Organization of Information and Systems
[a)] chanh@ism.ac.jp
[b)] kminami@ism.ac.jp

© 2026 Information Processing Society of Japan

This paper is work in progress and not peer-reviewed.

2. Related Work

Several studies have attempted to provide performance comparisons of the existing k-anonymization algorithms. However, they did not considered relating the results of information loss to performance of data analysis tasks like classification. For example, [13] compared several algorithms based on some utility metrics for estimating information loss, but did not relate them to any data analysis tasks. The study in [14] compared several algorithms on several datasets but only based on classification performance as the sole evaluation criterion and did not consider relating to it to information loss. Differently, [15] focused on the maximum distance to average vector (MDAV) algorithm and examined the correlation of the utility metric called "Distortion" to classification performance on several datasets. However, the study could not identify any clear or consistent correlation between the two. In light to determining a good utility metric for predicting classification performance, our previous works [7], [8] showed that showed that the data utility metric NCP provided a nice correlation with classification performance when considering the k-anonymization algorithms using local recoding approach. However, inconsistency was found when considering generalization algorithms; even with excessive information loss, anonymous data provided by these algorithms on ADULT dataset only suffered from minor deterioration in classification results.

On the other hand, existing work also considered the characteristics of original data as an input for anonymization to provide "useful" anonymous data. The work in [11] revised the state-of-the-art Mondrian [5] by incorporating the entropy of each QID attribute to properly decide the QID attribute (a.k.a., data feature) to be partitioned. On each partitioning step, the features with higher entropy will be prioritized. In fact, high-entropy features are often considered less important in classifiers as they are more random and correlate more poorly with the target variable [16]. Similarly, the work in [12] utilized the normalized mutual information as a measurement of feature importance, and generalized each feature such that its importance was maximized. Both works demonstrated substantial improvement in classification performance. Inspired by these works, in this paper, we explore the possibility of utilizing the feature importance as a data utility metric, in addition to NCP, and investigate whether it is well-correlated with classification performance.

3. Study Design

This section describes the metrics, algorithms, datasets, and experimental settings used in this paper. Section 3.1 presents the metrics for quantifying information loss (NCP) and classification performance (F_1 score). Section 3.2 describes the method for estimating feature importance based on permutation. The considered algorithms and datasets are presented in Section 3.3 and Section 3.4, respectively. Finally, Section 3.5 describes the experimental settings.

3.1 Metrics

Existing works on data anonymization often utilize a data utility metric to quantify the degree of information loss incurred on the anonymous data. In this paper, the Normalized Certainty Penalty (NCP) [6] is considered. NCP applies value-wise penalty (with respect to QIDs) and normalizes it by the number of records and the number of QIDs. It is calculated as Eq. (1).

$$NCP = \frac{1}{|\mathbb{Q}|} * \sum_{i}^{|\mathbb{Q}|} \mathcal{P}_i \tag{1}$$

where $\mathbb{Q} = \{Q_1 \dots Q_{|\mathbb{Q}|}\}$ is the set of QID attributes a.k.a. the *data features*. $\mathcal{P}_i$ is the *per-feature penalty* of feature Q_i, normalized by the number of records $|D|$.

For the numerical QID attribute/feature, Let $\Delta N_i(X)$ be the min-max range of the domain of Q_i bounded by the multi-set X. Intuitively, when a numerical value $Q_i(r_j)$ of record r_j on feature Q_i is anonymized, it becomes ambiguous in the min-max range of values in its equivalence class E. As a result, $\mathcal{P}_i = \frac{\Delta N_i(E)}{\Delta N_i(D)}$.

For categorical QIDs, when a value is anonymized by generalization, it becomes ambiguous among its siblings under the same ancestor. Let $\Delta C_i(X)$ be the cardinality of the values of Q_i bounded by the multi-set X, and $\mathcal{L}(A_i(r_j))$ be the number of leaf nodes under the ancestor $A_i(r_j)$ of r_j, $\mathcal{P}_i = \frac{\mathcal{L}(A_i(r_j))}{\Delta C_i(D)}$. In case of local recoding, existing works do not discuss the use of NCP where the mode value $\overline{E^c}$ is chosen as the representative value for its records. In this paper, we set $\mathcal{P}_i = 1$ if $Q_i(r_j) \neq \overline{E^c}$ and 0 otherwise.

For classification, F_1 score is a popular measurement to assess how correctly a classification model predicts the target variable, which comprised in both precision and recall for an overall assessment. F_1 score, denoted as S in this paper, is calculated by the number of true positives (TP), false positives (FP), and false negatives (FN) as Eq. (2).

$$\begin{gathered} S = 2\,\frac{Precision * Recall}{Precision + Recall}, \\ s.t.\ Precision = \frac{TP}{TP+FP},\ Recall = \frac{TP}{TP+FN} \end{gathered} \tag{2}$$

3.2 Permutation Feature Importance

This work employs the permutation feature importance (PFI) [17] to assess the impact of each QID attribute on the overall classification performance. Let q_i be the value vector of the i-th QID attribute Q_i of D. The process to determine the importance PFI_i of every q_i is outlined by Algorithm 1.

Firstly, values in q_i is shuffled to break the association between itself and the target variable. This creates the data $\hat{D}_i$ with the permuted $\hat{q_i}$ in place of q_i. Then, the considered classifier f is trained on $\hat{D}_i$ to provide predictions, which is evaluated against the truth values to get the F_1 score S_i; we denote this whole process as $S_i = f\{\hat{D}_i\}$ for simplicity. Finally, the importance PFI_i of Q_i in D is calculated as the difference between S_i and the baseline $S_0 = f\{D\}$, i.e., the F_1 score when no QID attribute is permuted. The more

© 2026 Information Processing Society of Japan

Algorithm 1: Permutation Feature Importance

Input: Data D (or anonymous data D'), Classifier f, Value vector q_i (or anonymous value vector q'_i)
Output: PFI_i

$S_0 \leftarrow f\{D\}$; // get F_1 score before permutation
$\hat{q_i} \leftarrow \text{SHUFFLE}\{q_i\}$;
$\hat{D_i} \leftarrow \{D \setminus q_i\} \cup \hat{q_i}$; // replace q_i with permuted $\hat{q_i}$
$S_i \leftarrow f\{\hat{D_i}\}$; // get F_1 score after permuting q_i
return $PFI_i \leftarrow S_0 - S_i$;

distant PFI_i is from 0, the more importance Q_i is. Note that Algorithm 1 can also be used for determining the importance of an anonymous QID attribute Q'_i in anonymous data D'.

3.3 Algorithms

This paper considers four popular k-anonymization algorithms to generate anonymous data.

Datafly [2] is the first ever k-anonymization proposed in the literature. It utilizes the full-domain generalization approach. Anonymization is performed by generalizing the entire value domain(s) of one or many QID attribute(s), based on the pre-defined generalization hierarchies. Typically, generalizing a QID attribute to a higher generalization level makes its value become more general and less specific, resulting in a smaller set of distinct possible values, which helps achieve k-anonymity. Datafly applies a greedy iterative heuristic that generalizes the QID with the largest number of distinct values at each iteration until k-anonymity is achieved.

Incognito [3] is also a popular full-domain generalization algorithm. Incognito examines all possible combinations of generalization levels of all QID attributes. It constructs a generalization lattice, where each node represents a solution (i.e., a combination of generalization levels), and performs a bottom-up breadth-first search to find a set of solutions satisfying k-anonymity. Then, the optimal solution is selected from such a set by the practitioners with respect to their specific goal(s). In this paper, we selected the one with the lowest NCP as it minimizes information loss incurred on the data.

Mondrian[5] is a popular local recoding algorithm that utilizes the domain partitioning approach. It firstly chooses the QID attribute with the widest range of values and partition the data into two groups on its median. Then, the partitioning procedure is repeated recursively on each resulting group, until no partition can produce two groups of size at least k. Anonymization is applied to the records within each group by assigning an identical representative value for each QID attribute. In this paper, we selected the most frequent value for each categorical QID attribute, and the arithmetic mean for each numerical QID attribute. This results in several data groups of size $\geq k$ whose records are identical (with respect to QID); the anonymous data is simply a concatenation of these anonymous groups, thus guaranteeing k-anonymity.

K-Member [4] is also a popular local recoding algorithm but utilizes a clustering approach. For each cluster, it firstly picks an initial record (seed) from the data, then selects $k-1$ other records such that the information loss of that cluster is minimal. The seed is chosen randomly (only for the first cluster) or based on the furthest distance to the previous seed. The clustering process is repeated until there are less than k records remaining in the data. Finally, each remaining record is added to one of the previously-created clusters that results in the minimum information loss. Similar to Mondrian, the records of each cluster are anonymized by assigning an identical representative value for each QID attribute. We also used the most frequent value for the categorical QID attribute, and the mean value for the numerical QID attribute.

3.4 Datasets and Data Preparation

In this work, two datasets were used for evaluation, the details of which are summarized as follows:

- **ADULT** [10]: This is a well-known dataset that has been used in almost any existing works on data anonymization. Thus, it becomes the de facto choice. This work chose eight attributes as QID, namely `sex`, `age`, `race`, `marital-status`, `education`, `native-country`, `workclass`, and `occupation`. Among them, `age` was a numerical QID attribute, while the others were categorical QID attributes. The target attribute was `salary-class`. Any non-QID attribute was excluded. Finally, the records with missing values were removed, resulting in 30,162 records.
- **UK Housing Prices** [9] (**HOUSING** for short): This dataset contains prices paid for real-estate property in the United Kingdom since 1995. At the time of acquired, the data had more than 30 million records. For reducing computational complexity, we filtered the data to include only records for London county in 2002. The selected QID attributes were `month` (derived from *date*), `district`, `street`, and `is_new`. The target was `price-class`, which was derived from *price* by binarizing it into "$>$*200k*" and "$\leq$*200k*." All QID attributes were considered categorical, and non-QID attributes were excluded. After removing missing values, the data consisted of 35,309 records.

The data preparation procedure was similar to the one in our previous work [8]. For each dataset, we filtered the data to keep only the unique records, from which we sampled the same number of records for every value of the sensitive attribute to create a balanced data. Then, we applied a stratified 80/20 split on the balanced data to create the *sample data* and *validation data*, respectively. Such a procedure resulted in the sample/validation pair of 4,800/1,200 records for ADULT and 10,400/2,600 records for HOUSING dataset. Finally, the sample data was used as input for the anonymization algorithms to create corresponding *anonymous data*, while the validation data was kept in the original form and used as the test set for classification.

© 2026 Information Processing Society of Japan

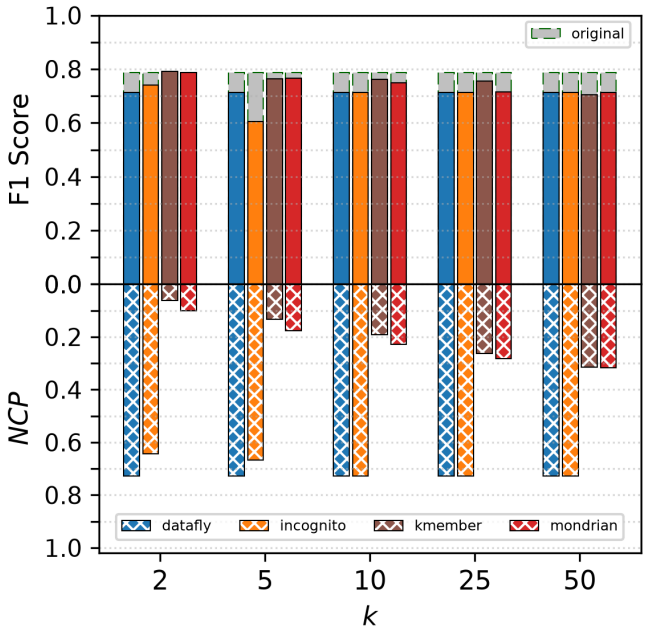


Fig. 1: NCP and F_1 score of the evaluated methods on ADULT dataset

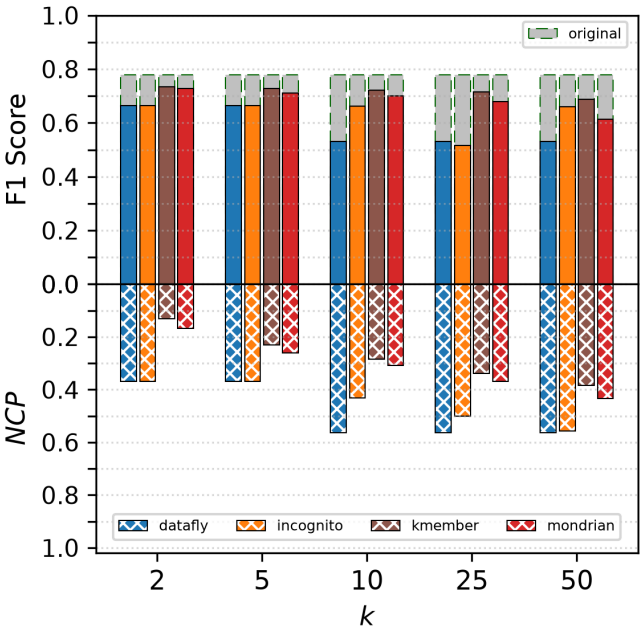


Fig. 2: NCP and F_1 score of the evaluated methods on HOUSING dataset

3.5 Experimental Settings

The experiment in this paper was conducted using Python 3.13 on a MacBook Pro with 32GB of RAM, running macOS Tahoe on Apple M4 chip. The k-anonymity values were varied from 2 to 50. For classification, we employed the Histogram-Based Gradient Boosting implementation from the public library `skicit-learn` [18]. For PFI, we repeated Algorithm 1 ten times for every QID attribute and reported the average value.

4. Results

Figure 1 and 2 show the NCP and F_1 scores of anonymous data generated by the evaluated k-anonymization algorithms on ADULT and HOUSING dataset, respectively. The gray bars show the F_1 score of the original data (i.e., the baseline), the bars with plain color on the upper half show the F_1 scores of anonymous data, and the bars with cross pattern on the lower half shows the NCP. Also, Table 1 and 2 summarize the feature importance PFI and the per-feature penalties $\mathcal{P}$ of each QID attribute of ADULT and HOUSING dataset, respectively. The gray-out cells in both tables are those QID attributes that were suppressed after anonymization.

On ADULT dataset (Fig. 1), Datafly and Incognito resulted in severe information loss (high NCP) across all values of k, while those of KMember and Mondrian were much lower and increased gradually following k. Such results suggested that classification performance on anonymous data generated by Datafly and Incognito would be more seriously harmed. However, it was revealed that their F_1 scores only dropped slightly compared to the baseline, except for Incognito at $k = 5$. Also, all four algorithms resulted in relatively comparable F_1 scores.

According to Table 1, `marital-status`, `education`, `age`, and `occupation` were found to be the significantly more important features, among which `marital-status` completely stood out. While Datafly and Incognito ended up suppressing most QID attributes, which explained the excessively high NCP, `marital-status` usually remained in the anonymous data. Interestingly, the only case where `marital-status` was also suppressed (Incognito, $k = 5$), F_1 score dropped significantly. On the other hand, KMember and Mondrian did not suppress any QID attributes, hence the low NCP. Among the four important features, `marital-status` had the lowest $\mathcal{P}$ when anonymized by KMember and Mondrian.

On HOUSING dataset, Fig. 2 also shows that Datafly and Incognito caused higher NCP than KMember and Mondrian across all values of k. However, the gaps among them were smaller compared to ADULT dataset. Yet, this time, it is shown that Datafly and Incognito resulted in clearly lower F_1 scores, especially for Datafly. Additionally, compared to ADULT dataset, Datafly and Incognito had much lower NCP, but resulted in the larger gaps in F_1 scores compared to the baseline.

Table 2 shows that `district` and `street` were the only important features. `street` was always suppressed by Datafly and Incognito, and always had the highest $\mathcal{P}$ when anonymized by KMember and Mondrian, reaching almost 100% as k increased. When both `district` and `street` were suppressed (e.g., Datafly with $k \in \{10, 25, 50\}$ or Incognito with $k = 25$), F_1 scores dropped drastically.

5. Discussion

Based on the results in Section 4, Datafly and Incognito caused severe NCP on ADULT dataset, but managed to retain the most important feature `marital-status`. Thus, the

© 2026 Information Processing Society of Japan

Table 1: Feature importance and per-feature penalty on ADULT dataset

Metric	Method	Feature (QID Attribute)							
		sex	age	race	marital-status	education	native-country	workclass	occupation
PFI	Original	0.007	**0.037**	0.000	**0.112**	**0.059**	0.008	0.005	**0.035**
$\mathcal{P}$ $k=2$	Datafly	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	Incognito	0.000	0.260	1.000	0.484	0.412	1.000	1.000	1.000
	KMember	0.003	0.102	0.017	0.044	0.118	0.085	0.059	0.065
	Mondrian	0.058	0.065	0.027	0.042	0.198	0.139	0.089	0.192
$\mathcal{P}$ $k=5$	Datafly	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	Incognito	0.000	1.000	0.000	1.000	1.000	1.000	1.000	0.335
	KMember	0.007	0.272	0.031	0.096	0.265	0.128	0.111	0.151
	Mondrian	0.109	0.143	0.050	0.099	0.334	0.163	0.166	0.356
$\mathcal{P}$ $k=10$	Datafly	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	Incognito	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	KMember	0.007	0.411	0.027	0.141	0.396	0.144	0.159	0.252
	Mondrian	0.139	0.198	0.073	0.144	0.426	0.169	0.221	0.463
$\mathcal{P}$ $k=25$	Datafly	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	Incognito	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	KMember	0.008	0.539	0.024	0.196	0.535	0.161	0.254	0.381
	Mondrian	0.166	0.277	0.091	0.191	0.508	0.174	0.289	0.566
$\mathcal{P}$ $k=50$	Datafly	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	Incognito	0.000	1.000	1.000	0.484	1.000	1.000	1.000	0.335
	KMember	0.003	0.643	0.033	0.254	0.603	0.172	0.326	0.480
	Mondrian	0.204	0.351	0.101	0.220	0.533	0.177	0.332	0.619

Table 2: Feature importance and per-feature penalty on HOUSING dataset

Metric	Method	Feature (QID Attribute)			
		month	district	street	is_new
PFI	Original	0.009	**0.143**	**0.113**	0.000
$\mathcal{P}$ $k=2$	Datafly	0.250	0.220	1.000	0.000
	Incognito	0.250	0.220	1.000	0.000
	KMember	0.176	0.009	0.332	0.005
	Mondrian	0.030	0.065	0.577	0.000
$\mathcal{P}$ $k=5$	Datafly	0.250	0.220	1.000	0.000
	Incognito	0.250	0.220	1.000	0.000
	KMember	0.228	0.022	0.666	0.004
	Mondrian	0.069	0.139	0.835	0.000
$\mathcal{P}$ $k=10$	Datafly	0.250	1.000	1.000	0.000
	Incognito	0.500	0.220	1.000	0.000
	KMember	0.267	0.038	0.830	0.002
	Mondrian	0.114	0.208	0.914	0.000
$\mathcal{P}$ $k=25$	Datafly	0.250	1.000	1.000	0.000
	Incognito	0.000	1.000	1.000	0.000
	KMember	0.365	0.064	0.924	0.001
	Mondrian	0.183	0.337	0.958	0.000
$\mathcal{P}$ $k=50$	Datafly	0.250	1.000	1.000	0.000
	Incognito	0.000	0.220	1.000	1.000
	KMember	0.480	0.096	0.957	0.002
	Mondrian	0.297	0.449	0.974	0.013

F_1 scores of their resulting anonymous data were not seriously harmed. On the contrary, despite the lower NCP on HOUSING dataset, Datafly and Incognito eventually suppressed all important features, hence the lower F_1 scores. For KMember and Mondrian, on both datasets, no suppression was done and information loss on each feature was low. Therefore, the F_1 scores usually did not decreased largely.

Such results suggested that, even when the overall information loss on the anonymous data, estimated by NCP, was high, if the specific loss on the important features were low, classification performance of models trained on such data would not be affected significantly. Therefore, feature importance should be considered in addition to NCP to estimate the impact of anonymization on classification performance. For instance, while NCP assumes a unified weight $\frac{1}{|\mathbb{Q}|}$ for every per-feature penalty $\mathcal{P}_i$, one should consider weighting the contribution of each $\mathcal{P}_i$ based on PFI. Equation (3) shows an example PFI-based weighted NCP, denoted as NCP_{FI}.

$$NCP_{FI} = \sum_{i}^{|\mathbb{Q}|} \frac{PFI_i}{\sum PFI} \mathcal{P}i \tag{3}$$

$\frac{PFI_i}{\sum PFI}$ is the normalized PFI of Q_i, which serves as the weight of $\mathcal{P}_i$ in NCP_{FI}. Figure 3 and 4 compare the correlation between NCP and the degradation in F_1 scores (denoted as ΔS) with the correlation between NCP_{FI} and ΔS of all algorithms on ADULT and HOUSING datasets, respectively.

On both datasets, the correlation coefficient (R^2) improves significantly when considering NCP_{FI}. However, the correlation coefficient on ADULT dataset is still relatively low even with NCP_{FI}. Yet, the improvement introduced by NCP_{FI} suggests that incorporating feature importance into information loss estimation is promising for predicting the influence of anonymization on subsequent classification performance.

© 2026 Information Processing Society of Japan

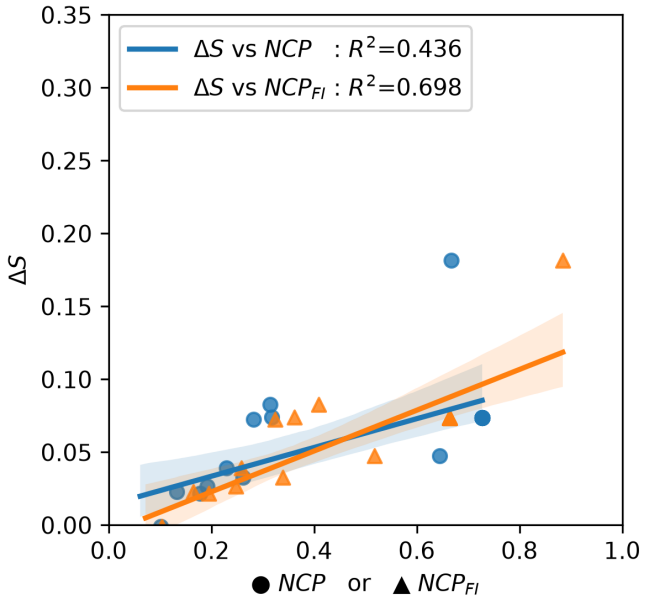


Fig. 3: Correlation between ΔS and NCP/NCP_{FI} of all algorithms on ADULT dataset

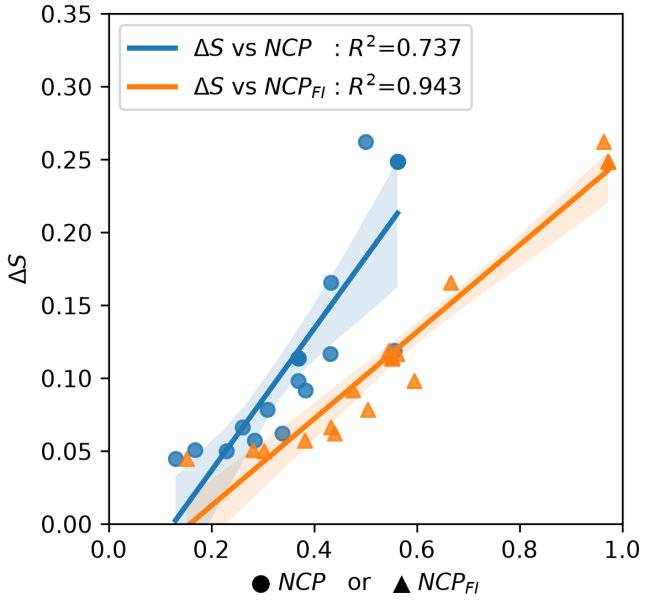


Fig. 4: Correlation between ΔS and NCP/NCP_{FI} of all algorithms on HOUSING dataset

6. Conclusion and Future Work

This paper investigate the influence of feature importance in addition to information loss incurred by anonymization on the performance of classification models trained on anonymous data. Experimental results suggest that information loss on important features deal more impact on classification performance. Essentially, even if the overall information loss, estimated by Normalized Certainty Penalty (NCP), is high, classification performance is not seriously affected if information loss on important features is low. Thus, utilizing feature importance as normalization weights for NCP potentially provides better prediction on classification performance. In the future work, a proper incorporation of feature importance into NCP will be investigated.

Acknowledgments This research was supported by JST Grant Number JPMJKP24U5, Japan. This research contains HM Land Registry data © Crown copyright and database right 2021. This data is licensed under the Open Government Licence v3.0.

References

[1] SWEENEY, L.: k-ANONYMITY: A MODEL FOR PROTECTING PRIVACY, *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, Vol. 10, No. 05, pp. 557–570 (online), DOI: 10.1142/S0218488502001648 (2002).

[2] Sweeney, L.: *Datafly: a system for providing anonymity in medical data*, pp. 356–381 (online), DOI: 10.1007/978-0-387-35285-5_22, Springer US (1998).

[3] LeFevre, K., DeWitt, D. J. and Ramakrishnan, R.: Incognito: efficient full-domain K-anonymity, *Proceedings of the 2005 ACM SIGMOD International Conference on Management of Data*, SIGMOD '05, New York, NY, USA, Association for Computing Machinery, p. 49–60 (online), DOI: 10.1145/1066157.1066164 (2005).

[4] Byun, J.-W., Kamra, A., Bertino, E. and Li, N.: Efficient k-anonymization using clustering techniques, *Proceedings of the 12th International Conference on Database Systems for Advanced Applications*, DASFAA'07, Berlin, Heidelberg, Springer-Verlag, pp. 188–200 (2007).

[5] DeWitt, D. J., LeFevre, K. and Ramakrishnan, R.: Mondrian Multidimensional K-Anonymity, *22nd International Conference on Data Engineering*, ICDE '06, Los Alamitos, CA, USA, IEEE Computer Society, p. 25 (online), DOI: 10.1109/ICDE.2006.101 (2006).

[6] Ghinita, G., Karras, P., Kalnis, P. and Mamoulis, N.: Fast data anonymization with low information loss, *Proceedings of the 33rd International Conference on Very Large Data Bases*, VLDB '07, VLDB Endowment, pp. 758–769 (2007).

[7] Chanh Minh TRAN and Kazuhiro MINAMI: Evaluation of Data Utility, Classification Influence, and Vulnerability of k-Anonymization Algorithms, *2026 Symposium on Cryptography and Information Security*, SCIS 2026, (online), available from ⟨https://www.iwsec.org/scis/2026/program.html?20260126#3C1⟩ (2026).

[8] Tran, C. M., Yamaguchi, T. and Minami, K.: Evaluation of k-Anonymization Algorithms on Data Utility, Classification Performance, and Vulnerability on Multiple Datasets, 研究報告コンピュータセキュリティ *(CSEC)*, SIG Technical Reports, pp. 1–7 (2026).

[9] HM Land Registry: Price Paid Data, `https://www.gov.uk/government/statistical-data-sets/price-paid-data-downloads`.

[10] UCI Machine Learning: Adult Census Income, `https://www.kaggle.com/datasets/uciml/adult-census-income`.

[11] LeFevre, K., DeWitt, D. J. and Ramakrishnan, R.: Workload-aware anonymization, *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '06, New York, NY, USA, Association for Computing Machinery, p. 277–286 (online), DOI: 10.1145/1150402.1150435 (2006).

[12] Li, J., Liu, J., Baig, M. and Wong, R. C.-W.: Information based data anonymization for classification utility, *Data & Knowledge Engineering*, Vol. 70, No. 12, pp. 1030–1045 (online), DOI: https://doi.org/10.1016/j.datak.2011.07.001 (2011).

[13] Ayala-Rivera, V., McDonagh, P., Cerqueus, T. and Murphy, L.: A Systematic Comparison and Evaluation of k-Anonymization Algorithms for Practitioners, *Trans. Data Privacy*, Vol. 7, No. 3, pp. 337–370 (2014).

[14] Slijepčević, D., Henzl, M., Klausner, L. D., Dam, T., Kieseberg, P. and Zeppelzauer, M.: k-Anonymity in practice: How generalisation and suppression affect machine learning classifiers, *Computers & Security*, Vol. 111, p. 102488 (online), DOI: https://doi.org/10.1016/j.cose.2021.102488 (2021).

[15] RodríGuez-Hoyos, A., Estrada-JiméNez, J., Rebollo-Monedero, D., Parra-Arnau, J. and Forné, J.: Does

© 2026 Information Processing Society of Japan

k-Anonymous Microaggregation Affect Machine-Learned Macrotrends?, *IEEE Access*, Vol. 6, pp. 28258–28277 (online), DOI: 10.1109/ACCESS.2018.2834858 (2018).

[16] Quinlan, J.: *C4.5: Programs for Machine Learning*, Ebrary online, Morgan Kaufmann (2014).

[17] Fisher, A., Rudin, C. and Dominici, F.: All Models are Wrong, but Many are Useful: Learning a Variable's Importance by Studying an Entire Class of Prediction Models Simultaneously, *Journal of machine learning research : JMLR*, Vol. 20 (2019).

[18] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, E.: Scikit-learn: Machine Learning in Python, *Journal of Machine Learning Research*, Vol. 12, pp. 2825–2830 (2011).

© 2026 Information Processing Society of Japan